

$$4.2 \quad p(x) = a_4 x^4 + a_3 x^3 + a_2 x^2 + a_1 x + a_0$$

$$\vec{x} = \begin{pmatrix} a_4 \\ a_3 \\ a_2 \\ a_1 \\ a_0 \end{pmatrix} \quad \vec{b} = \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix} \quad A \vec{x} = \vec{b}$$

$$A = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 16 & 8 & 4 & 2 & 1 \end{pmatrix}$$

$$A^T = \begin{pmatrix} 0 & 1 & 16 \\ 0 & 1 & 8 \\ 0 & 1 & 4 \\ 0 & 1 & 2 \\ 1 & 1 & 1 \end{pmatrix}$$

$$A \cdot A^T = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 5 & 31 \\ 1 & 31 & 341 \end{pmatrix}$$

$$x = A^T \cdot (A \cdot A^T)^{-1} \cdot \vec{b}$$

$$(A \cdot A^T)^{-1} = \begin{pmatrix} 186/115 & -31/46 & 13/230 \\ -31/46 & 17/23 & -3/46 \\ 13/230 & -3/46 & 1/115 \end{pmatrix}$$

$$A^T \cdot (A \cdot A^T)^{-1} = \begin{pmatrix} 53/230 & -7/23 & 17/230 \\ -51/230 & 5/23 & 1/230 \\ -103/230 & 11/23 & -7/230 \\ -129/230 & 14/23 & -11/230 \\ 1 & 0 & 0 \end{pmatrix}$$

$$A^T \cdot (A \cdot A^T)^{-1} \cdot \vec{b} = \begin{pmatrix} -3/5 \\ 7/5 \\ 3/5 \\ 4/5 \\ 1 \end{pmatrix}$$